

Improving RAG for Vietnamese Legal Documents with Cross-Reference Resolution

Research Specification · version 2 · Sau quyết định: Reframed the unsupported gap about missing Vietnamese legal

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1. Problem statement

- **RAG pipelines degrade on Vietnamese legal text with cross-references** — Standard RAG treats each chunk independently and fails to follow references between legal provisions, causing incomplete context and incorrect answers.
- **Cross-references between Vietnamese legal texts are an explicit challenge** — The user explicitly highlights provisions with cross-references between documents as the special focus for RAG improvement.

2. Research questions

- **How can cross-references be automatically extracted from Vietnamese legal texts?** — What parsing and linking methods can identify references to other articles, clauses, or laws in Vietnamese legal language?
- **What retrieval architecture best incorporates cross-reference structure into RAG?** — Should retrieval use graph traversal, query expansion, or reranking to include referenced provisions alongside directly matched chunks?
- **How should end-to-end RAG be evaluated on cross-reference-heavy legal queries?** — What metrics and datasets can capture whether the system retrieves all necessary referenced articles and generates legally faithful answers?

3. Related-work matrix

Study	What it did	Feedback type	What is still missing	Source
SPIRIT 2013 explanation and elaboration: guidance for protocols of clinical trials (2013)	Provides guidance for clinical trial protocols with a checklist and explanation.	Unrelated domain	Does not address legal text retrieval or cross-reference resolution.	DOI
Landsat-8: Science and product vision for terrestrial global change research (2014)	Describes Landsat 8 capabilities for Earth observation and global change research.	Unrelated domain	No relevance to legal document retrieval or cross-reference resolution.	DOI

Study	What it did	Feedback type	What is still missing	Source
LexRank: Graph-based Lexical Centrality as Salience in Text Summarization (2004)	Introduces LexRank, a graph-based method for computing sentence salience in text summarization.	Graph-based text analysis	Focused on summarization, not on retrieval of cross-referenced legal provisions.	DOI
Low intensity transcranial electric stimulation: Safety, ethical, legal regulatory and application guidelines (2017)	Abstract not available from the provider.	Unavailable	Not assessable for relevance to legal RAG.	DOI
Analysis of Points of Interests Recommended for Leisure Walk Descriptions (2025)	Presents a dataset for studying advertisements in RAG using queries and document segments from MS MARCO V2.1.	RAG evaluation dataset	Not focused on legal domain or cross-reference resolution.	DOI
A high-resolution geospatial assessment of the rooftop solar photovoltaic potential in the European Union (2019)	Geospatial assessment of rooftop solar photovoltaic potential in the EU using satellite data and machine learning.	Unrelated domain	No relevance to legal text retrieval.	DOI
Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG (2025)	Surveys agentic RAG architectures, focusing on autonomous agents in retrieval pipelines.	RAG survey	Does not specifically address cross-reference resolution in Vietnamese legal texts.	DOI
Findings of the Association for Computational Linguistics: EMNLP 2023 (2023)	Abstract not available from the provider.	Proceedings	Not a single paper; no specific focus.	DOI
Natural Language Information Retrieval (1999)	Abstract not available from the provider.	Generic IR	Not specific to legal cross-references.	DOI
A Comprehensive Survey on Cross-modal Retrieval (2016)	Surveys cross-modal retrieval methods for retrieving data across different modalities.	Cross-modal retrieval survey	Focuses on different modalities, not on legal text cross-references.	DOI
A Survey of Graph Retrieval-Augmented Generation for Customized Large Language Models (2025)	Surveys GraphRAG, which uses knowledge graphs to enhance retrieval and generation for domain-specific LLMs.	GraphRAG survey	Does not address specific legal cross-reference resolution for Vietnamese.	DOI
PathRAG: Pruning Graph-based Retrieval Augmented Generation with Relational Paths (2025)	Proposes PathRAG, which retrieves key relational paths from indexing graphs to reduce redundancy and improve prompting.	GraphRAG method	Not tailored to legal cross-references; evaluated on general datasets.	DOI

Study	What it did	Feedback type	What is still missing	Source
Ask in Any Modality: A Comprehensive Survey on Multimodal Retrieval-Augmented Generation (2025)	Surveys multimodal RAG systems covering retrieval, fusion, augmentation, and generation across modalities.	Multimodal RAG survey	Focuses on multiple modalities, not on legal cross-references.	DOI
VideoRAG: Retrieval-Augmented Generation over Video Corpus (2025)	Introduces VideoRAG, a framework for retrieval-augmented generation over video corpora.	Video RAG	Not relevant to textual legal documents.	DOI
VDocRAG: Retrieval-Augmented Generation over Visually-Rich Documents (2025)	Proposes VDocRAG for RAG over visually rich documents, treating them as unified images.	Document RAG	Does not address cross-reference resolution in text-only Vietnamese legal documents.	DOI
Optimizing Legal Text Summarization Through Dynamic Retrieval-Augmented Generation and Domain-Specific Adaptation (2025)	Introduces Dynamic Legal RAG for legal summarization, using BM25 and domain-specific NER.	Legal RAG	Focuses on summarization, not cross-reference resolution.	DOI
Hierarchical Retrieval with Evidence Curation for Open-Domain Financial Question Answering on Standardized Documents (2025)	Proposes HiREC, a hierarchical retrieval with evidence curation for financial QA on standardized documents.	Domain-specific RAG	Not applied to legal texts or cross-references.	DOI
Extracting Information from Brazilian Legal Documents with Retrieval Augmented Generation (2024)	Proposes a workflow to assess RAG setups for extracting structured data from Brazilian legal documents.	Legal RAG extraction	Focused on information extraction, not cross-reference resolution, and for Brazilian law.	DOI
Multi-Head RAG: Solving Multi-Aspect Problems with LLMs (2024)	Introduces MRAG using multi-head attention for multi-aspect retrieval to handle complex queries.	Multi-aspect retrieval	Does not model explicit cross-reference links between legal provisions.	DOI
Optimizing Answer Generator in Vietnamese Legal Question Answering Systems Using Language Models (2025)	Constructs Vietnamese legal provisions and QA datasets, fine-tunes small LLMs for legal QA with RAG.	Vietnamese legal QA	Does not address cross-reference resolution between provisions.	DOI
Vietnamese Legal Information Retrieval in Question-Answering System (2024)	Improves Vietnamese legal retrieval with data processing, normalized RRF, and active retrieval reranking.	Vietnamese legal retrieval	Does not explicitly model cross-references between legal documents.	DOI

Study	What it did	Feedback type	What is still missing	Source
A Self-Reflection Mechanism for Reducing Hallucination in Vietnamese Legal Question Answering Systems (2026)	Adds a self-reflection mechanism to Graph-RAG for Vietnamese legal QA, reducing hallucination.	Vietnamese legal GraphRAG	Focuses on self-reflection, not on cross-reference resolution.	DOI
Integrating Knowledge Graph with Retrieval-Augmented Generation for Vietnamese Legal Question Answering (2025)	Integrates an ontology-driven KG with RAG for Vietnamese legal QA, encoding cross-reference relations.	Cross-reference KG	Does not address ambiguous or temporally changing cross-references, and limited to three law domains.	DOI

4. Research gap

- Current Vietnamese legal retrieval methods ignore explicit cross-references between provisions**
 - What prior work achieved:* Recent Vietnamese legal retrieval systems improve data processing, normalize Reciprocal Rank Fusion, and apply active reranking to boost retrieval performance on legal QA benchmarks (source: 8991a8c9).
 - Remaining limitation:* These systems treat legal documents as independent chunks and do not follow explicit cross-references such as 'as stipulated in Article X of Law Y', so retrieved context may omit mandatory provisions outside the directly matched chunk.
 - Why that limitation matters:* Legal answers often depend on multiple linked provisions; missing a referenced article leads to incomplete context and incorrect legal interpretations, undermining trust in automated legal assistance.
 - Experiment that would test it:* Create a set of Vietnamese legal queries whose answers require following one or more cross-references, and compare the proportion of necessary provisions retrieved by the current system against oracle expansion using cross-reference links.
- Cross-reference resolution in Vietnamese legal knowledge graphs does not handle temporal validity and ambiguous references**
 - What prior work achieved:* An ontology-driven knowledge graph for Vietnamese legal QA (source: b79e46ba) encodes cross-reference relations and uses graph-aware retrieval to provide article-level citations.
 - Remaining limitation:* That knowledge graph is restricted to three law domains and does not resolve references that are ambiguous (e.g., pointing to multiple possible articles) or that may become outdated when laws are amended or replaced.
 - Why that limitation matters:* Vietnamese laws are frequently revised; a system that cannot track whether a referenced provision is still in force risks citing repealed or superseded text, leading to legally invalid answers.
 - Experiment that would test it:* Construct a test set of cross-references with known temporal status (active, repealed, ambiguous) and measure the proportion of correctly resolved references by the existing graph vs a version-aware resolver.

5. Proposed approach

We propose a cross-reference-aware RAG pipeline for Vietnamese legal documents. First, a cross-reference graph builder parses statutes and extracts explicit references to articles, clauses, and laws, resolving them to nodes in a versioned knowledge graph. Second, a version-aware resolver assigns temporal status (active/repealed/ambiguous) to each reference. Third, a reference-aware retrieval module expands the query with linked provisions and reranks chunks by combining semantic similarity with graph proximity. Fourth, we construct a benchmark of cross-reference-heavy legal questions with annotations of the minimum set of required provisions, and evaluate retrieval recall and answer faithfulness against standard dense retrieval.

6. Expected contributions

- **Cross-reference graph builder for Vietnamese legal texts** — We define a parsing and linking component that extracts explicit statutory references and constructs a graph of nodes for laws, chapters, sections, articles, clauses, and points, with edges representing cross-reference relations.
- **Version-aware reference resolver** — We implement a resolver that assigns temporal status to cross-references by consulting consolidated and historical law texts, distinguishing active, repealed, and ambiguous references.
- **Reference-aware retrieval module** — We develop a retriever that uses the cross-reference graph to expand queries and rerank chunks based on graph proximity and semantic similarity, ensuring referenced provisions are included before generation.
- **Benchmark with cross-reference annotations** — We construct a dataset of Vietnamese legal questions annotated with the minimum set of provisions required to answer, including those reached through cross-references, enabling systematic evaluation of reference-aware retrieval.
- **Evaluation protocol for reference-aware retrieval** — We define metrics for retrieval recall and citation precision on cross-reference-heavy queries, along with an ablation plan to isolate the effect of reference expansion.

7. Claim–evidence matrix

- **The cross-reference graph builder extracts explicit statutory references with precision at least 0.90 and recall at least 0.85 on a manually annotated sample of 500 Vietnamese legal references**
 - *Claim*: The cross-reference graph builder extracts explicit statutory references with precision at least 0.90 and recall at least 0.85 on a manually annotated sample of 500 Vietnamese legal references.
 - *Baseline*: A rule-based regex extractor over the same legal corpus
 - *Metric*: Precision and recall of reference extraction on the annotated sample
 - *Evidence*: Manual annotations by two legal experts with adjudication; evaluation on held-out references not used during development
 - *Refutation condition*: Precision falls below 0.90 or recall falls below 0.85 on the held-out sample.
 - *Cited evidence*: **(no source attached)**

- **Reference-aware retrieval using cross-reference expansion improves Recall@10 by at least 10 points over dense retrieval without expansion on cross-reference-heavy Vietnamese legal**
 - *Claim*: Reference-aware retrieval using cross-reference expansion improves Recall@10 by at least 10 points over dense retrieval without expansion on cross-reference-heavy Vietnamese legal queries.
 - *Baseline*: Dense retrieval with a multilingual sentence transformer fine-tuned on Vietnamese legal text, without cross-reference expansion
 - *Metric*: Recall@10 on the cross-reference subset of the newly constructed benchmark
 - *Evidence*: Mean Recall@10 over the same 300 cross-reference queries, three random seeds for any training components
 - *Refutation condition*: Mean Recall@10 of the reference-aware method does not exceed the dense baseline by at least 10 points, or the improvement disappears when expansion is ablated.
 - *Cited evidence*: UNSUPPORTED — Integrating Knowledge Graph with Retrieval-Augmented Generation for Vietnamese Legal Question Answering (2025)
- **The version-aware resolver assigns correct temporal status (active, repealed, ambiguous) to cross-references with accuracy at least 0.80 on a held-out set of 200 annotated referenc**
 - *Claim*: The version-aware resolver assigns correct temporal status (active, repealed, ambiguous) to cross-references with accuracy at least 0.80 on a held-out set of 200 annotated references.
 - *Baseline*: A non-versioned resolver that uses only current law text and ignores amendment history
 - *Metric*: Classification accuracy on temporal status of references in the held-out set
 - *Evidence*: Annotation of temporal status by legal experts using consolidated and historical law texts
 - *Refutation condition*: Accuracy falls below 0.80 on the held-out set, or the version-aware component does not improve over the non-versioned baseline by at least 5 points accuracy.
 - *Cited evidence*: (no source attached)
- **Providing reference-expanded context to the generator increases citation precision in generated legal answers by at least 15 points over baseline RAG without expansion.**
 - *Claim*: Providing reference-expanded context to the generator increases citation precision in generated legal answers by at least 15 points over baseline RAG without expansion.
 - *Baseline*: Standard RAG pipeline with dense retrieval and a fine-tuned Vietnamese legal LLM, without cross-reference expansion
 - *Metric*: Citation precision of generated answers on the cross-reference benchmark, judged by legal experts
 - *Evidence*: Human evaluation of 200 generated answers for whether each cited provision supports the answer
 - *Refutation condition*: Citation precision does not exceed the baseline by at least 15 points, or the increase is not significant under a paired bootstrap test at $p < 0.05$.
 - *Cited evidence*: UNSUPPORTED — A Self-Reflection Mechanism for Reducing Hallucination in Vietnamese Legal Question Answering Systems (2026)

8. Experimental protocol

- **TN1 — Reference extraction precision and recall benchmark**
 - Compare the proposed cross-reference graph builder against a rule-based regex extractor
 - On a manually annotated sample of 500 Vietnamese legal references held out from development
 - Metric: precision and recall of extracted statutory references, reported as mean over three annotation adjudication rounds
 - Success: precision ≥ 0.90 and recall ≥ 0.85 on the held-out sample
 - *Tests claim*: The cross-reference graph builder extracts explicit statutory references with precision at least 0.90 and recall at least 0.85 on a manually annotated sample of 500 Vietnamese legal references
- **TN2 — Temporal status classification accuracy**
 - Compare the version-aware resolver with a non-versioned resolver that uses only current law text
 - On a held-out set of 200 references annotated by legal experts with temporal statuses
 - Metric: classification accuracy for active, repealed, and ambiguous statuses
 - Success: accuracy ≥ 0.80 and exceeds the non-versioned baseline by at least 5 points
 - *Tests claim*: The version-aware resolver assigns correct temporal status (active, repealed, ambiguous) to cross-references with accuracy at least 0.80 on a held-out set of 200 annotated references
- **TN3 — Reference-aware retrieval Recall@10**
 - Compare reference-aware retrieval (with cross-reference expansion and graph-reranking) against dense retrieval without expansion
 - On 300 cross-reference-heavy Vietnamese legal queries from the newly constructed benchmark
 - Metric: Recall@10, mean and standard deviation across three random seeds
 - Success: mean Recall@10 of reference-aware method exceeds the dense baseline by at least 10 points
 - *Tests claim*: Reference-aware retrieval using cross-reference expansion improves Recall@10 by at least 10 points over dense retrieval without expansion on cross-reference-heavy Vietnamese legal queries
- **TN4 — Citation precision with reference-expanded context**
 - Compare a standard RAG pipeline (dense retrieval + fine-tuned Vietnamese legal LLM) against the same pipeline but with reference-expanded context
 - On 200 generated answers from the cross-reference benchmark, judged by legal experts
 - Metric: citation precision — proportion of cited provisions that truly support the answer
 - Success: citation precision improves by at least 15 points and the increase is significant under a paired bootstrap test at $p < 0.05$; generation is run three times with different seeds to estimate variance
 - *Tests claim*: Providing reference-expanded context to the generator increases citation precision in generated legal answers by at least 15 points over baseline RAG without expansion.

9. Baselines and metrics

Baselines: BM25 (Anserini defaults) for retrieval; dense retrieval (multilingual sentence transformer fine-tuned on Vietnamese legal text, e.g., paraphrase-multilingual-MiniLM-L12-v2); rule-based regex extractor for reference extraction; non-versioned resolver (current law only); standard RAG with fine-tuned PhoGPT-7B. Metrics: precision, recall, accuracy, Recall@10, nDCG@10, citation precision.

10. Ablation plan

Remove cross-reference expansion from the retrieval module while keeping graph-reranking: threatens claim 2 and claim 4 if recall or citation precision drops. Remove graph-proximity reranking but keep query expansion: threatens claim 2 if recall decreases. Replace the cross-reference graph builder with the regex extractor: threatens claim 1 if precision/recall falls below thresholds. Remove the version-aware historical consultation from the resolver: threatens claim 3 if accuracy drops below baseline difference.

11. Compute budget

- VRAM: **8.15 GB** (fits a 24 GB RTX 3090)
- Wall-clock: **10.67–33.33 h**
- Tokens: **7,680,000**
- API cost: **~\$2.42**

12. Risks and limitations

The held-out reference sample is drawn from the same Vietnamese law corpus as development, so leakage of document-specific patterns may inflate extraction metrics; we partition by laws to mitigate. The dense retrieval baseline may be insufficiently tuned (e.g., single learning rate) and thus underperform, exaggerating the improvement of our method. Human expert annotations are limited and may have subjectivity; we report inter-annotator agreement. We use a separate development split for all tuning decisions; no model is trained on the held-out evaluation sets. All experiments use a single random seed for training except retrieval which uses three; generation uses three seeds as described, but other components may still lack multi-seed robustness.

13. Open issues

- **Is there a public benchmark with Vietnamese legal cross-reference annotations for evaluating reference-aware RAG?** — Is there an existing public benchmark that annotates cross-reference chains and the minimum set of linked provisions needed to answer a Vietnamese legal query? If no such benchmark exists, what would be required to build one?
- **How to handle temporal validity of cross-references** — Laws are amended and replaced; a cross-reference may point to a provision that is no longer in force. How should the system maintain version-aware links?
- **How to disambiguate ambiguous legal references** — Some references point to multiple articles or use vague language; resolution may require semantic understanding beyond string matching.

14. Decision history

- 2026-08-16 09:14 · S1 · **A** — Truy hồi (retrieval)
- 2026-08-16 09:14 · S1 · **** — *(not applied)*
- 2026-08-16 09:14 · S1 · **** — *(not applied)*
- 2026-08-16 09:17 · S2 · **** — *(not applied)*
- 2026-08-16 09:34 · S4 · **A** — Sửa theo đề xuất